

Dynamic Charging and Path Planning for UAV-Powered Rechargeable WSNs Using Multi-Agent Deep Reinforcement Learning

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Abstract—Unmanned Aerial Vehicle (UAV)-powered 5G/6G networks integrated with rechargeable wireless sensor networks (RWSNs) offer promising solutions for extending system lifetime, collecting data, and providing computing services and power to sensor nodes (SNs). UAVs offer significant advantages, including exceptional mobility, cost-effective deployment, and the ability to be easily reprogrammed for a wide range of missions. However, the limited onboard power capacity of UAVs, coupled with the lack of dynamic and intelligent charging station (CS) management and inefficient path planning, can lead to SN failure in dynamic mobile environments. To address these challenges, we propose an energy-efficient laser-charged UAV (LCU)-enabled RWSN environment, wherein UAVs, powered by laser beams from ground-based stations, provide services, collect data, and transfer energy to SNs. We formulate a joint optimization problem involving power allocation, dynamic charging strategy (DCS), and path planning to minimize task completion time and sensor node death time. Given the NP-hard nature of the problem, we employ a stochastic game model based on a Markov decision process (MDP) for its solution. To solve this problem, we propose a deep reinforcement learning (DRL) based algorithm that enables real-time charging scheduling decisions while optimizing network performance. We introduce a multi-agent double deep Q-network (MA-DDQN) model to determine the optimal trajectories for all UAVs in large and complex environments. Simulation results demonstrate that the MA-DDQN approach outperforms state-of-

the-art techniques, showing significant improvements in terms of average delay, energy consumption, and task completion time.

Note to Practitioners—UAVs integrated with laser charging and deep reinforcement learning (DRL) enhance the performance and longevity of RWSNs in dynamic environments. Using energy-efficient LCUs for data collection and SN charging, this framework reduces task completion times, prevents SN failures, and optimizes energy use. The MA-DDQN algorithm, which optimizes UAV trajectories and charging schedules in real-time, offers a scalable and adaptive solution that outperforms traditional methods in energy consumption, data delivery efficiency, and system robustness. This approach benefits applications of smart agriculture by improving network reliability, extending sensor lifetimes, and enhancing system efficiency in challenging environments.

Index Terms—Double deep Q network, laser-charged UAV, rechargeable WSN, task completion time, trajectory planning.

I. INTRODUCTION

RECHARGEABLE wireless sensor networks (RWSNs) are having a significant impact across various industries, including civil, governmental, and military sectors, particularly within the frameworks of fifth-generation (5G) and beyond fifth-generation (B5G) technologies. In recent years, WSNs have been deployed in smart agricultural systems to optimize the control and management of agricultural production. These networks are capable of sensing environmental data and communicating with other WSN nodes to enhance the quality of agricultural outputs [1]. Recently, there has been growing interest from both industry and government in drone-mounted base stations (DBSs) due to their diverse applications. DBSs can be deployed to provide communication services for short-term use cases, such as in disaster-stricken areas or locations with high demand. Compared to traditional base stations, DBSs offer increased flexibility, mobility, and cost efficiency, making them a valuable addition to 5G and beyond networks [2].

In the agricultural application scenario, sensor nodes connected to the laser-charged UAV (LCU)-enabled RWSNs can collect agricultural data from multiple points, including temperature, humidity, and salinity. The existing literature on RWSN-based smart agriculture systems often deploys multiple sensors that communicate their data to a central server via a sink node or a UAV cluster head. However, due to the limited power of these sensors, ensuring reliable multi-

Received 27 September 2024; revised 6 January 2025; accepted 24 March 2025. Date of publication 10 April 2025; date of current version 30 May 2025. This article was recommended for publication by Associate Editor A. Pietrabissa and Editor K. Liu upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 62373255 and Grant 62003217, in part by the Natural Science Foundation of Guangdong Province under Grant 2024A1515011204, in part by the Shenzhen Natural Science Fund through the Stable Support Plan Program under Grant 20220809175803001, and in part by the Open Fund of the National Engineering Laboratory for Big Data System Computing Technology under Grant SZU-BDSC-OF2024-15. (Corresponding author: Xiaoshan Bai.)

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Digital Object Identifier 10.1109/TASE.2025.3558945

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hop transmission in a large-scale RWSN can be challenging. Additionally, the physical distance between the sensors and the limited battery life of these nodes further restrict the number and optimal distribution of the SNs [3]. Moreover, many sensor nodes are integrated with mobile vehicles to form WSNs for tasks such as wireless charging and data collection [4]. The primary wireless charging techniques employed in RWSNs are strongly coupled magnetic resonance and radio frequency (RF) waves [5]. These technologies enable targeted energy delivery based on the specific requirements of the sensor nodes, thereby enhancing the stability and controllability of the network's energy supply.

To address these challenges, LCU-enabled RWSNs have been increasingly deployed as a solution. Multiple autonomous UAVs can collaborate to sense the geographic environment and transmit their collected data to the designated location [6]. For instance, one study explored downlink communications between a laser-powered UAV and a ground-based base station [7]. Additionally, Jiang et al. [8] proposed a method to optimize the data transfer rate between drone-mounted base stations (DBS) and ground users by leveraging radio frequency charging through the DBS. Jaafer et al. [9] investigated the dynamics of battery charging for laser-powered drones, analyzing various charging methods in a range of settings. Similarly, the authors of [10] examined different charging techniques for drone networks, including those that utilize laser power.

The benefits of a laser-enabled UAV-based RWSN can improve real-world factors, such as weather conditions, which significantly impact laser transmission and overall system performance. These benefits include: 1) **Weather Adaptability:** Rechargeable wireless sensor networks can continuously monitor and collect data on weather conditions, such as temperature, humidity, and wind speed. This information can then be used to adjust the flight paths or power levels of the laser-charged UAVs to optimize performance in varying weather conditions [11]. 2) **Predictive Maintenance:** By monitoring environmental factors that could affect laser transmissions and overall system performance, such as extreme temperatures or high winds, UAVs can be proactively maintained to ensure they remain in optimal operating condition [12]. 3) **Dynamic Power Management:** Real-time data from the wireless sensor networks enables dynamic power management for the UAVs' laser charging systems. For example, if solar panel efficiency drops due to cloudy weather, the system can automatically adjust power consumption or allocate additional resources as needed. 4) **Enhanced Safety Measures:** By continuously monitoring environmental factors and adjusting operations accordingly, laser-powered UAVs can operate more safely in adverse weather conditions, reducing the risk of accidents or equipment damage. 5) **Increased Reliability and Efficiency:** With real-time data on weather conditions and system performance from the WSN, operators can make more informed decisions about when and how to deploy laser-charged UAVs for optimal results, minimizing downtime due to unforeseen environmental challenges. Overall, the integration of RWSNs with laser-powered UAVs has the potential to significantly improve their adaptability, reliability, and safety in varying weather conditions while maximizing overall system efficiency [13], [14].

Due to its excellent maneuverability, good channel conditions can be achieved by flying the UAV to a position with a high probability of line-of-sight (LoS) with ground users. However, the short battery life of a UAV limits the services it can provide. Long cables significantly increase the cost of deploying UAVs. The cable limits the range of the UAV and reduces its maneuverability. Wireless charging, such as RF and laser charging, is an attractive replacement for wired charging. RF charging can only operate over a minimal distance and has a large divergence angle to charge multiple devices. To address the key challenge of inefficiency in UAV trajectory planning and charging station (CS) management causing UAV collisions and battery failure, laser-powered UAV-based RWSN systems provide a suitable and sustainable power source [15], [16]. The effectiveness of charged UAV-based RWSNs is constrained by several challenges during flight and hover. These are outlined below:

- **Limited battery life:** RWSNs may face battery limitations, especially in large-scale deployments requiring long-distance data transmission. This can be addressed by optimizing power consumption and implementing efficient power management strategies.
- **Inefficient charging strategies:** Charging is limited by the range of laser beams, requiring the UAV to stay within a specific distance from the charging station. This can be mitigated by deploying multiple charging stations or using UAVs with longer flight ranges. Atmospheric conditions like fog, rain, or dust can also attenuate laser beams, reducing charging efficiency. Using higher-power lasers or adaptive beam steering can help minimize these effects.
- **Laser communication limitations:** Environmental factors such as weather, atmospheric disturbances, and line-of-sight obstacles can affect laser communication. Our approach addresses these challenges by using adaptive beam steering and advanced optical technologies for reliable data transmission.

To address the challenges faced by UAVs, recent research has introduced innovative approaches leveraging artificial intelligence (AI) and machine learning (ML) techniques, such as reinforcement learning (RL), deep Q-networks (DQN), and DRL. These advanced methods empower autonomous UAVs to explore their environments, make independent decisions, and optimize operations in real-time. The study in [17] explored model-free Q-learning (QL) and RL approaches for autonomous UAV mission planning. These methods enable UAVs to navigate through constrained environments and obstacles, selecting the optimal trajectory for data collection missions. Similarly, ML techniques like DQN have been extensively utilized to optimize trajectory planning, thereby enhancing UAV performance, extending flight times, and broadening their application range. The primary objective of DQN is to help UAVs identify the best course of action to maximize system payoffs. Building on these methods, some studies have integrated Double DQN (DDQN) techniques to approximate Q-functions. This approach leverages neural networks' capacity to model complex and dynamic environments [18], [19], [20]. These works also incorporate effective path planning algorithms and speed control mechanisms to ensure UAVs meet recharge requirements efficiently. Another

study proposed an energy-efficient deep learning-based architecture that enables UAVs to collaboratively collect data from distributed low-level sensors while simultaneously charging their batteries at multiple randomly deployed CSs [21]. Wireless energy transfer systems for recharging UAVs during operation have also been investigated in [22] and [23]. These studies formulated UAV energy and data transfer optimization as a Markov Decision Process (MDP) framework, solving it using the multi-agent Double Deep Q-Network (MA-DDQN) model. Despite these advancements, there remains a notable gap in the literature regarding the joint optimization of UAV power management, dynamic charging strategies (DCS), and path planning strategies. Addressing these interconnected challenges could unlock further efficiency and reliability in UAV operations.

Motivated by the aforementioned issues, we propose an energy-efficient LCU-enabled RWSN framework for smart agriculture, aiming to minimize task completion time and sensor node lifetimes by optimizing power, DCS deployment, and path planning while meeting data requirements [24]. The MA-DDQN method enhances decision-making and stability by using target and behavior networks. UAVs adjust routes to mitigate weather effects on laser transmission, ensuring optimal performance. Our approach integrates laser-charging technology, dynamic power management, and DRL-based path optimization, making it adaptable to real-world conditions and complex environments. We chose the MA-DDQN framework for 5G/6G networks due to its stability, scalability, and ability to handle dynamic, decentralized environments. Unlike DQN and Q-learning, MA-DDQN mitigates overestimation and ensures faster convergence. Its multi-agent design enables efficient coordination for resource allocation and energy management, outperforming MILP in terms of computational efficiency. Additionally, it adapts well to large-scale networks and handles delayed rewards better than K-means, making it ideal for complex 5G/6G scenarios. The main contributions of our work can be summarized as follows:

- We present an energy-efficient LCU-enabled RWSNs framework for smart agriculture, where multiple LCU-enabled RWSNs are deployed as an aerial base station (ABS) to form a user association, collect data from SN, and send the data to the ground control system to minimize task completion time and SN death time during data collection.
- We formulate the joint optimization problem of power optimization, DCS, and path planning as an MDP model. To solve the optimization problem mentioned earlier, we propose a DRL framework based on the MA-DDQN algorithm to learn the trajectory of the laser-charged UAV in a large, complex environment. To solve the non-convex optimization model while optimizing the multiple trajectories, the MA-DDQN algorithm is integrated with a Q-learning technique that takes into account that the UAV hovers at a fixed location to operate the SNs during the data collection mission and selects the best trajectory.
- We perform extensive simulations to assess the performance of our proposed MA-DDQN scheme, comparing it to state-of-the-art approaches in terms of average

delay, energy consumption, and task completion time. The simulation results show that MA-DDQN outperforms benchmark methods such as DQN, Q-learning, MILP, and K-means. Specifically, the MA-DDQN algorithm outperforms the benchmark schemes by reducing task completion time by 25%, reducing energy consumption by 30%, increasing data delivery efficiency by 27.25%, and extending SN battery life by 40% through optimized charging. These improvements are particularly significant in dynamic, real-world conditions, demonstrating the framework's potential for large-scale applications like smart agriculture.

The rest of this paper is arranged as follows: Section II presents a review of related works. Section III provides a description of the proposed system model. In section IV, the optimization problem is discussed. The proposed solution is presented in V. The performance evaluation is presented in section VI. A conclusion is presented in section VII.

II. RELATED WORK

In this section, we review recent advancements in data collection using laser-charged UAVs, explore strategies for minimizing task completion time with UAVs, and examine optimization techniques for trajectory planning. These topics highlight the evolving role of UAVs in improving operational efficiency, enhancing energy management, and addressing challenges in dynamic and complex environments.

The comparison between our work and the existing works is listed in Table I.

A. Laser-Charged Uav-Assisted Data Collection

For drones to serve effectively as mobile base stations or chargers, they require significantly extended operating times. As a result, intensive research has been conducted on wireless charging methods that can sustain UAVs during longer missions, with a particular focus on solar and laser energy solutions [34], [35]. While solar energy is a viable option, its functionality is limited to daylight hours. Laser beam transmitters, on the other hand, provide a continuous energy source, enabling UAVs to operate for extended periods regardless of the time of day. An innovative Internet of Things (IoT) system integrating UAVs, IoT devices, and laser beam drones (LBDs) was proposed by Lahmeri et al. [36]. In this system, UAVs collect data from IoT devices while LBDs supply power to the UAVs via laser beams, ensuring uninterrupted operation. Similarly, Zhao and colleagues in [37] developed a network optimization framework that jointly optimizes the transmission power of the source, UAV, and LBD, as well as the UAV's trajectory in a mobile relaying system using laser-based charging. Their approach aims to maximize the weighted sum of energy efficiency, with the LBD recharging the UAV by emitting focused laser beams. Further advancements were made by Ranjha and Kaddoum in [15], who studied the joint optimization of resource allocation, trajectory design, and energy harvesting for ultra-reliable and low-latency communication (URLLC) UAV relay systems. Operating in decode-and-forward (DF) mode, these systems

TABLE I
COMPARISON OF OUR PROPOSED WORK WITH EXISTING STUDIES

Reference	Scenario	Method	Problem	Objective	Agent
[25]	Improving the energy utilization rate of UAV	IA-DRL	Multi-UAV path planning,	Reduce energy loss	Multiple UAV
[26]	Freshness of the information received	K-means,DQN	Device energy consumption, flying time	Minimize AoI	multiple UAV, IoT Clusters
[27]	Integrated task assignment for multi-agent and pickup delivery	Large neighbourhood search	Multi agent path-finding, Task assignment	Minimize the robots total travel delay	Multiple robots
[28]	Multi-robot task assignment strategy	two group distributed auction	Assignment package of groups	Minimize the robots total travel time	Multiple robots, heterogeneous vehicles
[29]	PCHDP for drones strategy	Heuristic algorithms	Package delivery, task assignment	Minimize the robots total travel time	Heterogeneous drones
[30]	UAV-enabled MEC	Successive convex approximation	Computational offloading, Resource allocation, Trajectory path	Minimize energy consumption and latency of UAVs	Single UAV, IoT devices
[31]	UAV-equipped wireless power chargers	Contract theory-based optimization	Terminal devices charging from the flying UAVs	Maximize the utility of energy server	Multiple UAV, terminal devices
[32]	O-RAN-enabled multi-UAV-assisted WSNs for smart agricultural system	MADRL	Task scheduling, Resource allocation, Trajectory path	Minimize energy consumption and network latency	Multiple UAVs, IoT devices, BSs
[33]	Laser-Charged UAV relying network	PPC Model	minimize power consumption of CS	Trajectory path, Charging power	Single UAV, GBS, UE
Proposed	Energy-efficient LCU-enabled RWSN framework	MA-DDQN	Minimize task completion time and the SN's death time	UAV power, DCS, and path planning optimization	Laser-charged UAVs, SN

enable a UAV to serve as a relay while being recharged mid-mission by a laser transmitter.

B. Trajectory Planning

Recently, there has been growing interest in optimizing trajectory planning for drone-based data acquisition. Various optimization techniques have been employed in existing approaches to enhance efficiency and performance. For instance, the works in [38] and [39] focus on minimizing task completion time in multi-UAV systems, addressing the challenges of coordinating multiple UAVs for efficient operation. In [40], the problem of age-optimal data acquisition in UAV-enabled WSNs is investigated, where a UAV is utilized to gather data from ground sensor nodes. Similarly, the study in [41] examines UAV-based data collection in real-time RWSNs, emphasizing the need for high information freshness. In this work, multiple UAVs are deployed to ensure timely and efficient data acquisition.

C. Task Completion Time

In [38], the problem of minimizing task completion time in a multi-UAV system is addressed by simultaneously optimizing the UAVs' positions, speeds for data collection, and the transmit power of the sensors. These optimizations are performed under load requirements and energy constraints. Building on this, the study in [42] employs a two-step method to optimize the number and placement of cluster heads that aggregate data from sensors, determining efficient UAV trajectories. Additionally, a heuristic genetic algorithm is proposed to achieve a low-complexity, suboptimal solution. However, the work in [38] does not account for the effects of small-scale fading, which could influence the results. To extend UAV network lifetimes, the authors of [43] introduce a non-disruptive RWSN

model. This model enables wireless recharging of UAVs without requiring them to visit charging platforms. Moreover, the WRUN model schedules the Real-Time Wireless Sensor Network (RWSN) to activate and release energy exclusively during designated charging periods, enhancing overall energy efficiency.

D. Optimization Strategies

In [44], a novel optimization combining Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) was proposed to extend the lifetime of UAV-assisted cluster-based WSNs in 3D environments, addressing coverage and data collection limitations [45], [46]. Studies explored IoT, the Internet of Medical Things (IoMT), Micro-aerial Vehicle (MAV) navigation algorithms, multi-UAV systems, and heuristic methods [47]. Energy efficiency was enhanced using MPC for eco-driving [48], while task assignment under constraints was tackled with local search techniques [49]. In [50], sensory-constrained quadrotors dynamically planned obstacle-avoidance paths using receding horizon-based methods. The long short-term memory (LSTM) and DRL were combined for multi-objective optimization, demonstrating UAV deployment with LoRaWAN gateways in rural IoT networks [51]. A study [52] analyzed ABS placement in user hotspots for offloading traffic from terrestrial base stations. MA-DDQN algorithms were applied to UAV trajectory optimization and laser charging. MEC systems with UAVs utilized UAV agility for efficient computing [18], while RL techniques were used for IoT data collection [53]. CNN-DQN combinations enabled multi-robot network path planning, and DRL-based methods improved multi-UAV path planning for wireless data collection [54]. UAVs with charging and monitoring systems were

suggested to overcome battery and power limitations in remote areas.

On the other hand, the proposed MA-DDQN method offers significant advantages over classical optimization techniques, such as those in Grieco et al. [47], particularly in dynamic, large-scale UAV-RWSN environments. Unlike classical approaches that rely on precomputed schedules and centralized algorithms—which struggle with scalability and real-time adaptability—MA-DDQN leverages distributed reinforcement learning to enable autonomous, real-time decision-making. It dynamically optimizes UAV trajectories, power allocation, and charging strategies while accounting for uncertainty in sensor node energy demands and environmental changes. This adaptability minimizes task completion time and energy waste, outperforming static classical methods that often assume fixed topologies or deterministic parameters. Additionally, MA-DDQN's multi-agent framework efficiently scales to complex networks, whereas classical optimization becomes computationally prohibitive for large, mobile systems. While traditional methods provide interpretable, optimal solutions for small-scale static problems, MA-DDQN excels in dynamic scenarios where flexibility and real-time coordination are critical.

Based on the literature mentioned above, this paper proposes a new network framework that includes DCS based on LCU, power, and path planning. The UAV is deployed as a mobile collector to collect data from sensors and as a laser-charged vehicle to replenish its energy. We investigate the duration of task completion in such a network by simultaneously optimizing path planning, laser UBS deployment, and energy optimization during a data collection mission. The scalability of our proposed MA-DDQN approach addresses the challenges of large-scale agricultural environments with heterogeneous SN deployments. It ensures effective and efficient operation in complex, diverse settings through the following methods: a) Improved Communication Protocols: We develop robust protocols for seamless communication between heterogeneous SNs and multi-UAVs, optimizing data transfer and coordination. b) UAV Coordination: The MA-DDQN algorithm coordinates multiple UAVs for efficient data collection, monitoring, and task management. c) Autonomous Operation: The system enables the autonomous operation of both SNs and UAVs, reducing the need for constant human supervision. d) Dynamic Resource Allocation: Dynamic resource allocation techniques optimize power, bandwidth, and processing capacity across SNs and UAVs. e) Scalable Infrastructure: A scalable infrastructure is designed to handle increasing nodes and UAVs while ensuring efficient operation. f) Energy Efficiency: The method focuses on energy-efficient solutions, extending the operational lifespan of SNs and UAVs without frequent recharging or refueling.

Moreover, some of the key notations and descriptions are listed in Table II.

III. SYSTEM MODEL

In this section, we discuss the modeling of network, charging, and energy in RWSNs enabled by LCUs deployed for smart agriculture applications. The energy-efficient LCU-enabled RWSN, as depicted in Fig. 1, consists of key

TABLE II
SUMMARY OF KEY NOTATION

Notation	Description
D	The distance of UAV in a given region
PLoS	The probability of data transmission loss
NLoS	The probability of Non-data transmission loss
$\bar{B}$	The capacity of battery for UAV
T	The total time that all UAVs are required to complete the assigned tasks
D_f	The drag force of the UAV depends on the airspeed
R_u	The circle radius that centered at the projection of UAV
P_s	Probability of transmission for an SN in one frame

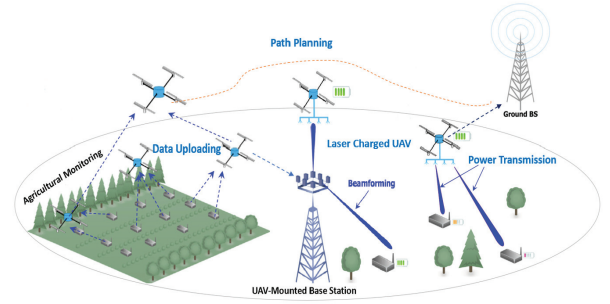


Fig. 1. Laser-Charged UAV data collection and rechargeable SN for agricultural monitoring.

components working together in this model: a laser beam, multiple sensor nodes (SNs), and a UAV-mounted base station. This model considers two key outcomes: (1) UAVs interact directly with mobile SNs to gather environmental data, and (2) SNs transmit collected data to multiple UAVs for real-time transmission. Using uplink ground-to-air (G2A) channels, UAVs cooperatively collect data, determining their flight paths and coverage areas while avoiding interference during communication. The 3D location coordinates of UAV $u \in \mathcal{U} = \{1, 2, \dots, U\}$ and sensor nodes $\kappa \in \mathcal{K} = \{1, 2, \dots, K\}$ are defined as (x_u, y_u, z_u) and $(x_\kappa, y_\kappa, 0)$, respectively. To collect data from the RWSN large-scale zones, the UAVs can follow a trajectory path and enter the communication range of the SNs, as shown in Fig. 1. Each of the τ equal time slots that make up this work cycle has a duration of $\Delta = T/\tau$. The time slots are expressed as $T_\tau = \{1, \dots, t, \dots, \tau\}$, where t is the index of the time slot that is now in effect.

SNs are low-power devices incapable of collecting all data from a given area. To address this, multiple LCUs are employed to maximize network lifetime and optimize power usage. These LCUs collect data from the SNs and transmit it to a central base station. Since UAVs have limited battery capacity, they must visit a CS to recharge using laser beams within the time frame defined by the variable Δ . During operations, UAVs fly to the target area (TA) to gather data and return to the CS as needed for recharging.

A. Network Model

In this subsection, we define the operational area for LCU-enabled RWSNs to conduct their missions. Equipped with communication interfaces, these UAVs independently perform distributed data collection from a specified number of SNs within the designated TA. They can assess the status of SNs

and collect data using wireless communication protocols such as Zigbee, Bluetooth, or Wi-Fi. The communication quality between ground nodes and UAVs, as well as the UAVs' flight performance through wireless channels, is influenced by path loss (PL). The Gaussian fading in the air-to-ground (A2G) channel affects communication reliability between SNs and UAVs located at u . To model this, both line-of-sight (LoS) and non-line-of-sight (NLoS) conditions are considered using a shadow-free channel model to determine the probability of LoS communication between a UAV as ABS and a ground terminal. [55]. The probabilistic LoS is considered the communication path between the LCU u and the SN k at the location u , where the probability of loss can be expressed as [56]:

$$p^{los} = \frac{1}{1 + ae^{-b(\theta_k - a)}}, \quad (1)$$

where a and b indicate environmental characteristics particular to a certain location, such as rural, urban, or other terrains [57]. The angle of elevation between UBS and SN k is θ_k expressed as $\theta_k = \arctan\left(\frac{\bar{h}}{d_k}\right)$, where $\bar{h}$ is the height of ABS and d_k can be a horizontal distance between UAV u and SN k . The average path loss [58] between SNs κ and UAV u can be calculated as follows:

$$\Gamma u = P_{ku}^{LoS} l_{ku}^{LoS} + P_{ku}^{NLoS} l_{ku}^{NLoS}, \quad (2)$$

where $\mathbb{P}_{ku}^{LoS}$ denotes the LoS probability between the UAV u and the SN κ , $P_{ku}^{NLoS} = 1 - P_{ku}^{LoS}$ is NLoS probability. l_{ku}^{LoS} and l_{ku}^{NLoS} are the LoS and NLoS paths, which are defined as in [56] and [58].

B. Energy Consumption of Sensor

The RF power transmission concept is used from the UAV to the sensor. The free-space route loss model between the UAV and the sensor is what we use [59]. The power that the sensor κ got from the UAV during the time slot t can therefore be expressed as

$$P_\kappa = \xi \times h_\kappa^u \times p^r = \frac{\xi \rho_0 p^r}{D_t^\kappa}, \quad (3)$$

where the channel gain from UAV u to SN κ can be denoted as $h_\kappa^u = \frac{\rho_0}{D_t^\kappa}$. The variable ρ_0 can be denoted as the channel power of the distance reference $1m$, $D_t^\kappa = (x_t^u - x_t^\kappa)^2 + (y_t^u - y_t^\kappa)^2 + h^2$, indicates the squared Euler distance between the sensor κ and the UAV u , and $\xi \in (0, 1)$ is the efficiency of the conversion of SN energy from RF to DC [60]. To decide whether to use the UAV to charge sensor κ during time slot t , we define a binary variable called L_t^κ of the LCU that can charge SNs in a given time t . If $L_t^\kappa = 1$, it means that the SN κ gets charged by the UAV for the time duration L_t^{ch} ; otherwise, $L_t^\kappa = 0$. As sensors use their energy to monitor environmental data, their remaining energy at the start of the current time slot t can be stated as

$$E_t^\kappa = \begin{cases} E_0^\kappa - t \cdot \Delta \cdot r^\kappa + L_t^\kappa \cdot L_t^{ch} \cdot P_\kappa, & 0 \leq E_t^\kappa \leq E^f \\ 0, & E_t^\kappa \leq 0. \end{cases} \quad (4)$$

To indicate the total death time of sensor κ at time slot t , we then introduce the variable $T_d^\kappa[t]$, which may be written as

$$T_d^\kappa[t] = \begin{cases} t \cdot \Delta - t \frac{L_t^\kappa \cdot L_t^{ch} \cdot P_\kappa + E_0^\kappa}{r^\kappa}, & T_d^\kappa[t] > 0 \\ 0, & T_d^\kappa[t] < 0, \end{cases} \quad (5)$$

Using equations (4) and (5) as a foundation, we can calculate that the total death time of all sensors is equal to

$$T^{die} = \sum_{\kappa=1}^K T_d^\kappa \cdot \tau. \quad (6)$$

C. Assumption of Laser-Charged Power Model

We examine the free-air optical propagation between the charging station and the UAV. The power received from the UAV can be stated as [61]:

$$P_r = P_t C_t C_r \Upsilon_t v_e v_r P, \quad (7)$$

where P_t is the transmit power and C_t is the transmit gain of the charging station. The optical efficiency at the transmitter and receiver is given by v_t and v_r , respectively. The environmental attenuation parameter can be explained as $v_e = 10^{-\alpha L/10}$, where L is the distance between the UAV and the charging station. The environmental factor, denoted by α , is measured with the following values: 0.19 dB/km in clear weather, 6.9 dB/km in foggy conditions, and 28.9 dB/km in moderate fog [62].

$$P_r = P_t (D/\theta L)^2 v_t 10^{-\alpha L/10} v_r. \quad (8)$$

The loss of the free path can be denoted as $P = (\lambda/4\pi L)^2$, where λ is the wavelength of the laser. The receiver gain can be written as $C_r = (\pi D/\lambda)^2$, where D is the diameter of the receiver, $C_t = 16/\theta$ is the transmitter gain, and θ is the angle of divergence [63].

D. Uav Charging Model

The UAV uses lithium batteries for its power supply. When the battery is fully charged, its state of charge is C [64]. The state of charge of the battery is represented by the Coulomb count method as follows:

$$B^t = 1 - \frac{1}{C} \int_0^t I(\vartheta) d\vartheta, \quad (9)$$

where $I(\vartheta)$ is the amount of electricity charged by UAV at a given distance $d\vartheta$. The amount of battery that remains in the charge can be explained as follows [65]:

$$C^t = B^t C. \quad (10)$$

The remaining flight time of the UAV could be written in the following way;

$$f^t = \frac{B^t \varpi_m \varphi_e}{I^d}, \quad (11)$$

where φ_e is the battery discharge efficiency, φ_m is the motor efficiency, and I^d is the UAV's average working current. The UAV is expected to receive an average constant voltage from the laser energy it receives. The amount of energy stored in the battery during charging equals the initial charge minus the

charge consumed by the UAV plus the charge collected by the laser. During charging, the remaining flight time of the UAV is displayed as follows:

$$f_E^t = \frac{Q(t)\varphi_m\varphi_e}{I^d} + \frac{\frac{\varphi_m\varphi_e\varphi_c}{V} \int_0^t P_r(\vartheta)d\vartheta}{I^d}. \quad (12)$$

The variables in the equation are defined as follows: $Q(t)$ represents the battery capacity of the UAV at time t ; φ_m denotes the motor efficiency; φ_e represents the efficiency of the energy transfer system; I^d is the discharge current of the battery; φ_c indicates the charging efficiency; V is the voltage of the UAV's battery; and $P_r(\vartheta)$ represents the power received by the UAV at time ϑ , integrated over the charging period $[0, t]$. The first term represents the remaining flight time derived from the current battery capacity and system efficiencies. The second term reflects the energy contribution from charging during the specified period, adjusted for charging efficiency and other system parameters. Together, the equation illustrates the interplay between energy inputs, system efficiencies, and the UAV's operational time during the charging process.

E. Energy Consumption of Uav

The UAV uses energy for three purposes: a) propulsion energy (flying, hovering, and communication); and b) sending energy to the sensors.

- 1) Propulsion energy: UAVs use more energy when flying to collect data from SNs. The energy consumed by UAVs is classified into two types: In [66], communication-related energy is used in UAV communication systems to send and receive signals that are much smaller than propulsion energy and can be explained as:

$$P(u)^{fly} = P_0 \left(1 + \frac{3v^2}{U_{tip}^2} \right) + P_i \left(\sqrt{1 + \frac{v^4}{4v_0^2}} - \frac{v^2}{2v_0^2} \right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho\kappa Av^3, \quad (13)$$

where P_i is the induced power during the hovering operation and P_0 is the constant variable that represents the blade profile power. The trip speed of a rotor blade is U_{tip} . v_0 represents the main rotor that the UAV induces when it is hovering. Furthermore, d_u, κ, ρ , and A denote the fuselage drag ratio, the area of the rotor disc, the air density, and the reliability of the rotor, respectively [67]. Here is how to calculate the UAV's power consumption at $v = 0$, or hovering:

$$P_h = P_0 + P_i. \quad (14)$$

In our scenario, the UAVs employ communication-related power to gather data while hovering over SNs. Consequently, the power consumption of the UAV during the data collection phase is represented as follows:

$$P_{dc} = P_h + P_c. \quad (15)$$

We compute the UAV u 's total propulsion energy in a given time t as follows:

$$E_u^{cost} = P_{dc} + P(v)^{fly}. \quad (16)$$

- 2) Charging Sensors Energy: The energy used by the UAV during the time interval t to charge the sensors can be shown as

$$E_{ch}^u = \sum_{\kappa=1}^K \cdot L_t^\kappa \cdot L_t^{ch} \cdot p^r, \quad (17)$$

where L_t^κ and L_t^{ch} are parameters related to the sensor charging operation, and p^r denotes the charging power. The UAV's overall energy usage during the mission completion cycle is then:

$$E_u^{tot} = E_u^{cost} + E_{ch}^u + f_E^t. \quad (18)$$

IV. PROBLEM FORMULATION

In this section, we aim to minimize the UAV's task completion time and the SN's death time by jointly optimizing the power, DCS, and path planning based on energy and time constraints. For simplicity of our problem, we consider DCS of LCU-mounted as ABS $\{C = c_u[t] \in \{0, 1\}, \forall u, t\}$ [68], the power control of SNS $\{\beta = p_\kappa[t] \in \{0, 1\}, \forall \kappa, t\}$ [69], and the trajectory path can be $\{Q = q[n], \forall u, t\}$ [5], respectively. The formulation of this equation can be stated as:

$$\begin{aligned} \mathbf{P1} : \min_{Q, C, \beta} & \sum_{u=1}^U \sum_{\kappa=1}^K \sum_{j=1, j \neq \kappa}^K (\Psi f^t + T^{die}) \\ \text{s.t.} & \begin{cases} \mathbf{C1} : 0 \leq E_t^u \leq E_{laser}^u \\ \mathbf{C2} : \|u_{(laser)}\| \leq V_{max}, \forall \kappa(t) \in [0, T] \\ \mathbf{C3} : \sum_{u=1}^U \sum_{\kappa=1}^K x_{kj}^u = 1, \forall j = 1, \dots, \kappa, j \neq \kappa \\ \mathbf{C4} : \sum_{\kappa=1}^K x_{kp}^u - \sum_{j=1}^K x_{pj}^u = 1, \dots, K \\ \mathbf{C5} : \sum_{\kappa=1}^K x_j^u = 1, \forall u = 1, \dots, U \\ \mathbf{C6} : \text{Subtour elimination constraints.} \end{cases} \end{aligned} \quad (19)$$

The objective function (19) explains how to optimize power, DCS, and trajectory planning while working within various constraints from the LUC to the endpoint. We use the variable Ψf^t to minimize the UAV task completion cycle and T^{die} the SN death time by finding the best charging schedules for the UAV. The constraint **C1** ensures that the remaining energy of the UAV E_t^u is always greater than or equal to zero. The variable E_{laser}^u can show that when the UAV's energy becomes low, it sends a request to the LCU to replace its battery. The constraint **C2** specifies the maximum speed required for the UAV u to travel from CS to the target area in a given time t . The constraint **C3** states that each SN must be visited at least once by the UAV. The constraint **C4** states that when a UAV visits SNs, it must depart from the same CS location, the conservation of flow constraint. The constraint **C5** ensures that a UAV is used exactly once. Finally, the constraint **C6** states that the Miller Tucker Zemlin (MTZ)-based subtour elimination constraints, where tours not connected to the charging station under the same coverage area are omitted.

This makes solving the Mixed-Integer Nonlinear Programming (MINLP) problem in (19) particularly challenging. To address this, we divide the main problem into two subproblems: initially, the agent selects the optimal path to collect SN data, and then we employ the MA-DDQN framework to make rapid decisions during data collection in obstacle-filled environments. In the context of smart agriculture, UAVs cooperatively exchange vital monitoring information, such as soil moisture, temperature, plant health, and communication signals. This data is gathered from sensors and devices deployed across agricultural areas to monitor crop health and other farming operations. By working collaboratively in a distributed manner, UAVs can efficiently cover extensive agricultural areas and collect comprehensive data for analysis.

The primary goal is to enhance the efficiency and accuracy of the monitoring process. By sharing data and coordinating their movements, UAVs can reduce redundancy, cover larger areas, and provide a detailed, real-time overview of the agricultural landscape. This cooperative approach optimizes flight paths, prevents collisions, and ensures that all areas of interest are thoroughly monitored. Additionally, data sharing enables cross-validation of findings, increasing the overall reliability of the information collected. Our proposed approach leverages a decentralized control algorithm, allowing each UAV to make decisions based on local information, such as its position, speed, and the positions of neighboring UAVs. This decentralized mechanism supports real-time adjustments to avoid collisions, eliminating the need for centralized coordination and enabling efficient and adaptive operations.

V. THE PROPOSED SOLUTIONS

This section presents an MA-DDQN optimization algorithm to solve problem (19). The problem is first formulated as a MDP to provide a structured framework for the optimization. To address the multiple and complex state and action spaces inherent in smart agricultural systems, we model the joint power allocation, DCS, and trajectory optimization problem as an MDP. The proposed DRL-based technique allows UAV agents to resolve this multi-objective optimization problem effectively, enabling them to select the optimal routes for efficient data collection and operation.

We then provide a detailed explanation of the MA-DDQN algorithm. Problem 19 is a mixed-integer non-convex optimization problem, which makes finding the ideal solution a significant challenge. Our objective is to minimize the UAVs' task completion time and the sensor nodes' (SNs) death time by jointly optimizing power allocation and path planning within the LCU-enabled RWSNs. This optimization problem is reformulated as **P2**, which is expressed as follows:

$$\begin{aligned} \mathbf{P2} : \min_{Q, \beta} & \sum_{u=1}^U \sum_{\kappa=1}^K \sum_{j=1, j \neq \kappa}^K (\Psi f^t + T^{\text{die}}) \\ \text{s.t.} & : \mathbf{C2}, \mathbf{C3}, \mathbf{C4}, \mathbf{C5}. \end{aligned} \quad (20)$$

Here, the objective function (19) is to minimize the combined metrics of task completion time (Ψf^t) and SN death time (T^{die}) across all UAVs ($u \in U$) and SNs ($\kappa, j \in K$) while adhering to the specified constraints (**C2**, **C3**, **C4**, **C5**). This formulation

encapsulates the joint optimization of key parameters for effective operations in the network. As a result, the direct solution of the objective function (20) in **P2** is challenging. The path of multiple UAVs, where the agent tries to find the best path of the path during data collection. After decomposing the problem, we propose MDP frameworks to enable fast decisions in LCU-assisted RWSNs.

A. Markov Decision Process Model

This subsection defines Problem 20 as MDP since the UAV determines its charging decisions based on the network state at each time interval ($t \in \tau T$). According to the formulation of Problem 19, the MDP is represented as a tuple (S_t, A_t, R_t) , where S_t denotes the state space, A_t represents the action space, and R_t is the reward obtained by the agent during the time interval t . At each time step, the agent observes the current state S_t and selects an action A_t based on the environmental conditions to maximize the reward R_t . The execution of A_t transitions the system to a new state S_{t+1} . AI algorithms, particularly reinforcement learning methods, are well-suited to solving MDPs. In this context, MA-DDQN is employed to derive the optimal charging strategy for addressing Problem 19 [70].

Considering the complexity of managing numerous obstacles in three-dimensional smart agricultural environments, we adopt a two-stage control network approach as proposed in [67]. To achieve the goal of optimizing the UAV's trajectory, we solve Problem **P2**, reformulated as **P3**, which aims to minimize the task completion time and sensor node (SN) death time:

$$\begin{aligned} \mathbf{P3} : \min_{\mathbf{C}} & \sum_{u=1}^U \sum_{\kappa=1}^K \sum_{j=1, j \neq \kappa}^K (\Psi f^t + T^{\text{die}}) \\ \text{s.t.} & : \mathbf{C1}, \mathbf{C2}. \end{aligned} \quad (21)$$

In this framework, the UAV operates without prior knowledge of its surrounding environment. Using reinforcement learning principles, we transform the sensor node network into a digital model, facilitated by UAV assistance, to represent real-world applications in a computational form. The UAV's awareness is thus confined to local information, where its position determines its operational state and the conditions it encounters. This approach ensures that the UAV dynamically adapts to its environment, optimizing its trajectory while addressing challenges posed by complex agricultural settings and numerous obstacles.

B. Learning Framework for Uav-Assisted DCS in RWSNs

The state and action spaces for learning are discretized to simplify the problem and enable efficient training. The UAV-assisted RWSN trajectory and data-gathering process are formulated as follows:

- 1) Agent: The agent interacts with the environment to address the DCS problem by functioning as a laser charger, charging the SNs based on their requests. At each time slot t , the agent observes the current environment state S_t , takes an action A_t according to a given

policy, and transitions the environment to a new state S_{t+1} . The reward R_t corresponds to the agent's performance during this interaction. When training converges, the agent learns the optimal trajectory and charging strategy to maximize efficiency. This framework incorporates three critical elements: state space, action space, and reward functions.

- 2) **State Space:** In the DCS problem, the state space S_t comprises the following components:

- Sensor node death times (S_t^{die}): Represents the death times of all sensor nodes.

$$S_t^{\text{die}} = \{S_1^d, S_2^d, \dots, S_N^d\}, \quad (22)$$

where S_κ^d ($1 \leq \kappa \leq K$) is calculated as described in Equation (4).

- UAV residual energy (S_t^E): Denotes the remaining energy of the UAV for charging sensors.

$$S_t^E = \{S_t^e, u\}, \quad (23)$$

where S_t^e represents residual energy as explained in Equation (17).

- UAV position (S_t^p): Indicates the UAV's current position in its trajectory.

$$S_t^p = \{x_t^u, y_t^u, h\}. \quad (24)$$

Sensor positions are excluded from the state space as they remain constant during the mission.

- 3) **Action Space:** Actions in the DCS problem represent the UAV's decision to charge specific sensors. The action space A_t is defined as:

$$A_t = \{a_t^\kappa \mid 0 \leq t \leq \tau, 1 \leq \kappa \leq K\}, \quad (25)$$

where $a_t^\kappa = 1$: The UAV charges sensor κ at time t . $a_t^\kappa = 0$: The UAV does not charge sensor κ . After completing the action A_t , the system transitions to a new state, where

- Sensor node death time at $t + 1$:

$$T_{t+1}^\kappa = T_t^\kappa + a_t^\kappa \frac{P_d^t(t)B(t)}{\varsigma^\kappa} - \Delta, \quad (26)$$

where ς^κ is the energy consumption rate and Δ accounts for delays. UAV residual energy at $t + 1$:

$$E_{t+1}^u = E_t^u - E_{\text{total}}, \quad (27)$$

where E_{total} is the total energy consumed by the UAV during time slot t .

- UAV position at $t + 1$:

$$[x_{t+1}^u, y_{t+1}^u, h] = \begin{cases} [x_s^\kappa, y_s^\kappa, h], & \text{if } a_t^\kappa = 1, \\ [x_{t+1}^c, y_{t+1}^c, h], & \text{otherwise.} \end{cases} \quad (28)$$

- 4) **Reward Function:** After the agent takes an action A_t , the system receives a reward R_t , which depends on both the action A_t and the state S_t . The goal of the MDP is to maximize S_t by minimizing task completion time and sensor death time. The optimization objective (21) and the reward function are inversely related. To quantify

the advantage gained by the system from action A_t , we define the reward function as follows:

$$R_t = -\ln(\Psi f^t) - \ln(T^{\text{die}}), \quad (29)$$

where $T^{\text{die}} = \sum_{\kappa=1}^K T_d^\kappa \cdot \tau$, representing the cumulative sensor node death time overall sensors. $f^t = \frac{B_t \omega_m \varphi_e}{I_d}$, denoting the task completion factor.

The natural logarithm function is employed to standardize the values of f^t and T^{die} , ensuring that both components are normalized to the same order of magnitude. This standardization facilitates the successful integration of the two components in the reward computation. Conventional approaches struggle with the exponential growth of the state and action spaces over time, making it challenging to manage such complexity. However, the DDQN approach overcomes this limitation by using a neural network to approximate the Q-value function. This capability enables efficient handling of high-dimensional state and action spaces. Consequently, the 20 problem can be effectively resolved using an optimization technique based on MA-DDQN.

This framework enables efficient learning and optimization of UAV trajectories and charging strategies, ensuring reliable performance in RWSNs.

C. Multi-Agent-Based Double Deep Q-Learning Network Algorithm

For the DCS for the UAV to charge the sensors, we use the MA-DDQN algorithm, which is explained in detail in this section. The main goal is to handle the 20 problem in a way that minimizes both the UAV's task completion time and the sensors' death time, as demonstrated by Algorithm 1. The architecture of the MA-DDQN algorithm is shown in Fig. 3. Initially, we set the mini-batch δ , waiting iterations M , training frequency γ , learning rate α , memory pool size β , and update interval η . Next, we set the target network $Q'(\cdot)$ parameter $\phi' = \phi$ and randomly initialize the main network $Q(\cdot)$ parameter ϕ . To optimize the DDQN network's parameter ϕ , the method carries out $\Delta_{\max}$ iterations. The states S_1 and $s_t = 1$ are initialized before beginning each iteration: $1 \leq \Delta \leq \Delta_{\max}$. Afterward, the algorithm performs the subsequent actions for every time slot where $1 \leq t \leq \tau$. A random action with probability σ or an action that maximizes the value of $Q(S_t, A_t; \phi)$ with probability $1 - \sigma$ based on the state s_t is chosen by the main network $Q(\cdot)$. Next, to get the reward R_t , the UAV executes the action.

The following actions are taken to train and optimize the network if $t \geq M$ and t are a multiple of γ . Initially, we randomly use Ω samples from the memory pool β . To compute the Q-estimate $Q(s_g, A_g, \phi)$ for each sample (s_g, A_g, R_g, s_{g+1}) , we use the main network $Q(\cdot)$ and the target network $Q'(\cdot)$, respectively, which is $y_g = R_g + \Gamma Q'_m ax(s_g, A_g, \phi')$. To be executed during the next time slot, set $S_t = S_{t+1}$. When $t = \tau$ and $\sum_{n=1}^{\tau} s_t^{Lc} \geq \Delta$, we use $s_t = t$ to update the number of time slots and accumulate idle time. We also update $t = \tau + \frac{\sum_{n=1}^{\tau} s_t^{Lc}}{\Delta}$.

Algorithm 1 MA-DDQN Framework for UAV-Assisted Data Collection

- 1: **Initialize:** Learning rate α , discount factor γ , ϵ -greedy parameters ϵ , memory pool size β , and target network parameters $\phi' = \phi$.
- 2: Set the number of agents and their initial positions at the CS sites.
- 3: Initialize the main network $Q(\cdot)$ parameters ϕ and memory pool.
- 4: **for** each episode $\Delta = 1$ to $\Delta_{\max}$ **do**
- 5: Initialize the state S_1 for all agents.
- 6: **for** each time step $t = 1$ to τ **do**
- 7: Each agent selects an action A_t :
- 8: With probability ϵ , select a random action.
- 9: Otherwise, select $A_t = \arg \max Q(S_t, A; \phi)$.
- 10: Perform the action A_t and observe reward R_t and next state S_{t+1} .
- 11: Store the transition (S_t, A_t, R_t, S_{t+1}) in the memory pool.
- 12: **end for**
- 13: **for** each training step if $t \geq M$ and t is a multiple of γ **do**
- 14: Sample a mini-batch of Ω transitions (S_g, A_g, R_g, S_{g+1}) from the memory pool.
- 15: Compute target value $y_g = R_g + \gamma \max_{A'} Q'(S_{g+1}, A'; \phi')$.
- 16: Update the main network $Q(\cdot)$ by minimizing the loss:
- 17: **end for**
- 18: Update the target network parameters $\phi' \leftarrow \phi$ after every η steps.
- 19: At the end of the episode, agents return to their initial positions.
- 20: **end for**
- 21: **Output:** Optimized policies for all agents.

$$L(\phi) = \frac{1}{\Omega} \sum_{g=1}^{\Omega} (Q(S_g, A_g; \phi) - y_g)^2.$$

so that it can be used for accumulating idle time in the next iteration. The trend of the UAVs in the current slot, in addition to the data storage capacity and the remaining battery level, is taken into account by the service priority requirement. The service request priorities for SN κ for uploading data and charging at time slot t . The explanation of E_{κ}^{dt} and E_{κ}^{ch} is as follows:

$$E_{\kappa}^{dt} = \pi(t) \frac{b_{\kappa}^{dt}}{b_{\max}^{dt}}, \quad (30)$$

$$E_{\kappa}^{ch} = \mathfrak{I}_{\kappa}(t) \frac{b_{\max}^{ch} - b_{\kappa}^{dt}}{b_{\max}^{ch}}, \quad (31)$$

where b_{κ} is the power level of the SN to upload the data and $b_{\max}^{ch}$ is the maximum battery level of the UAV to recharge the SN in a given time t .

The learning rate, discount factor, and ϵ -greedy parameters are initialized in Algorithm 1 (lines 1–5), along with the number of agents starting the data collection mission at the CS

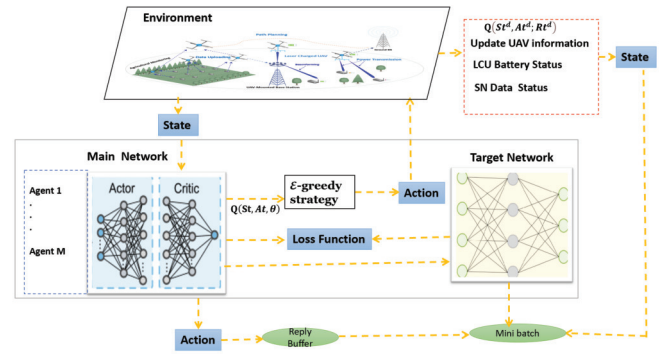


Fig. 2. MA-DDQN-based Network.

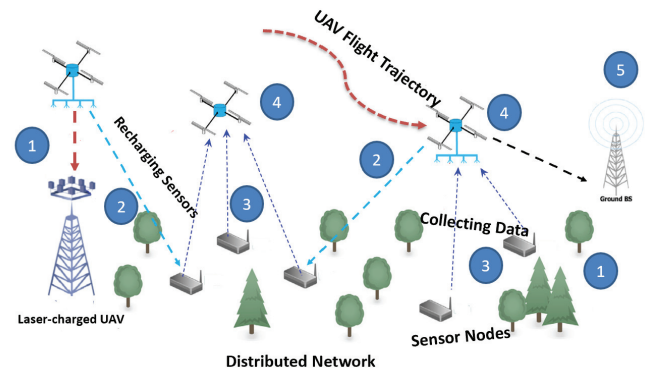


Fig. 3. Logical network topology diagram.

sites. Once the agents collect the observed data, they proceed to select their respective strategies. Lines 6–12 describe how each agent performs its activities: selecting an action, receiving a reward, transitioning to a new state, and interacting with the environment. During the training phase (lines 12–14), the sampled transitions of SN K are processed, and policy training is applied at each step. After making a final decision based on the selected samples, the agents complete the episode. They then return to the starting point to prepare for subsequent iterations, as outlined in lines 14–20.

Fig. 3 illustrates the logical network topology of the proposed approach, showcasing the communication and coordination between UAVs and SNs. The UAVs communicate seamlessly to recharge the SNs, optimize their flight paths, and ensure efficient coverage of the entire area of interest. The network topology diagram, incorporating the DDQN algorithm, is instrumental in organizing energy utilization and distribution, which significantly enhances path planning and DCS in RWSNs. The data collection process in a UAV-assisted RWSN-based smart agriculture system typically involves the following steps: 1. Laser Charging Technology: A ground-based laser system wirelessly transmits energy to UAVs in flight, enabling battery recharging without requiring the UAVs to land. This technology improves operational efficiency and extends flight duration. 2. Sensor Recharging: LCUs recharge multiple sensors within the agricultural monitoring area. 3.

Data Upload: The SNs upload data on key parameters such as soil moisture, temperature, and crop health. **4. Data Collection:** UAVs hover over the SNs and collect data from the WSN nodes. **5. Data Transmission:** Finally, the UAVs transmit the collected data to a central server for comprehensive analysis. This approach streamlines energy management, improves UAV efficiency, and facilitates robust data collection and processing in smart agriculture systems.

D. Complexity Analysis

In the context of LCUs, the proposed MA-DDQN approach effectively addresses the complexities of real-world conditions, such as varying weather. This framework enables multiple UAVs to learn and coordinate their actions to optimize energy efficiency while accounting for environmental factors. Each UAV functions as an independent agent, capable of navigating and recharging other UAVs, with the MA-DDQN algorithm trained in simulated environments that include diverse weather scenarios. By incorporating weather data into the state representation, the algorithm adapts to fluctuations in laser transmission strength, optimizing UAV trajectories and decision-making in real-time. This ability to integrate environmental data ensures that the UAVs learn to respond dynamically to changing conditions, enhancing overall system efficiency. The proposed MA-DDQN method is utilized for optimizing UAV trajectory paths, training performance within a reinforcement learning environment, and managing training episodes. When analyzing the complexity of reinforcement learning algorithms, it is essential to consider system constraints. Since input processing and output generation are computationally efficient, model execution operates in near real-time. For Q-value convergence to the ideal solution, specific conditions must be satisfied, as outlined in Algorithm 1. The computational complexity of the method for a fully connected LCU system is approximately $O\left(\sum_{u=1}^U \sum_{\kappa=1}^K (\Psi f^t + T^{\text{die}}) \cdot q_t\right)$, where κ represents the number of sensor nodes (SNs) in a given layer and q_t denotes the trajectory path at time t . The time required to train the DDQN method is estimated as $O\left(s \cdot \sum_{u=1}^U t_{l-q}\right)$, where s represents the number of states. Consequently, the minimization of task completion time and sensor deadline complexities in the MA-DDQN method can be expressed as $O\left[(A + s \cdot \sum_{u=1}^U \sum_{\kappa=1}^K (\Psi f^t + T^{\text{die}}) t_{q-1}]\right]$. Here, O represents the obstacle environment observed by the UAV agent, and A refers to the action space for decision-making.

VI. PERFORMANCE EVALUATION

This section evaluates the performance of DDQN frameworks with 3D obstacle environments against other conventional benchmark algorithms.

A. Simulation Setup

We present a smart agriculture monitoring system in a 4×4 km area, involving 10 UAVs with laser charging capabilities and 200 heterogeneous sensor nodes (SNs) monitoring parameters like soil moisture, temperature, and crop health.

The UAVs, operating at 100 meters and 10 m/s, act as both data collectors and chargers, transmitting data to a central base station. The goal is to minimize task completion time, enhance energy efficiency, and demonstrate scalability under dynamic conditions. Our MA-DDQN algorithm optimizes UAV trajectories, reduces redundancy, and prioritizes charging for SNs with low energy. Existing methods like DQN, Q-learning, MILP, and K-means struggle with challenges like trajectory planning, large-scale deployment, and real-time adaptability. DQN and Q-learning are inefficient in complex environments, while MILP is computationally infeasible for large-scale problems. K-means lacks dynamic resource allocation. In contrast, our MA-DDQN method leverages multi-agent reinforcement learning, adapting to real-world conditions such as fluctuating SN energy levels, obstacles, and adverse weather, overcoming the limitations of traditional methods. We used a Core i7 CPU running at 2.4 GHz and 16GB of RAM for training and testing operations, along with Python 3.7 and Tensorflow 2.0.

We compare the following proposed MA-DDQN schemes in multi-UAV assisted RWSNs with the following schemes, which can be listed as follows: (A) Deep Q-Network (DQN): is a technique for reinforcement learning that trains an agent to make decisions in a given environment by fusing Q-learning and deep learning [71]. (B) Q-learning: The UAV uses Q-learning to assist it in making decisions about where to gather data, how to best utilize the network, and how to enhance performance in this algorithm. The method by which the algorithm operates is the learning of a Q-function, a function that calculates the predicted benefit of a given action under a given condition [72]. (C) Mixed Integer Linear Programming (MILP): This optimization method aims to improve the performance and energy efficiency of WSNs by leveraging the capabilities of UAVs in a coordinated and optimized manner through mathematical programming techniques. It utilizes MILP techniques to optimize the operation of UAVs in a WSN [73]. (D) K-means: The method iteratively divides the data points into K clusters as each UAV collects data by reducing the variance within each cluster [74].

To assess the effectiveness of our proposed framework, we primarily evaluate the following metrics:

- 1) *The overall time taken to complete a UAV task: This is the total time the UAVs spend during the data collection mission and other processing of their mission-critical operations.*
- 2) *The Average Delay (AD): The AD metric estimates the average network delay from the source of the SNs to the UAV for all delivered packets. This can be expressed as: $A_d = \frac{P_k}{P_u}$. The sum of the packets generated by the SNs is P_k , while the sum of the packets delivered by the UAV is P_u .*
- 3) *Energy: The total energy that the UAV used to complete the assignment in the time allotted.*

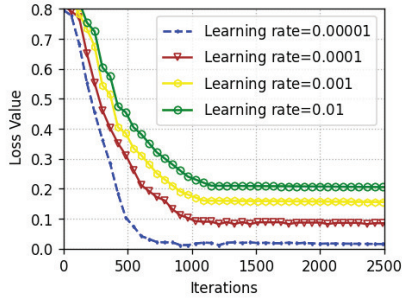
The simulations were carried out with the settings shown in Table III.

B. Convergence Analysis

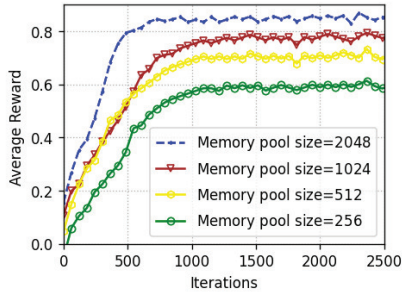
The investigation focuses on assessing the alignment of gathered data with ground truth and determining if further

TABLE III
SIMULATION PARAMETERS

Parameters	Values
Number of UAVs (U)	10
The UAV's weight (m_u, g) and 40N	
The overall surface area of rotor disks ($(\pi r_\rho^2 n \rho)$)	$0.20m^2$
Number of BS (χ)	1
maximum battery power level B_{level}	$4000mAh$
Distance between SNs (κ)	20m
The density of the air (ρ)	$1.225kg/m^3$
The speed of the UAV (v)	$10m/s$
Data transmission power consumption [77] P_{tx}^j	$0.1W$
S-curve parameters ($\in \beta$)	9.6, 0.16
Transmission with path loss (ηLoS)	1dB
Path loss for transmission with ($\eta NLoS$)	20dB
Density of noise (N_0)	-174dbm
The total bandwidth (B)	10MHz
Learning rate α	0.5
Discounting element factor (θ)	0.9
Weight parameters (W_1, W_2)	1, -100



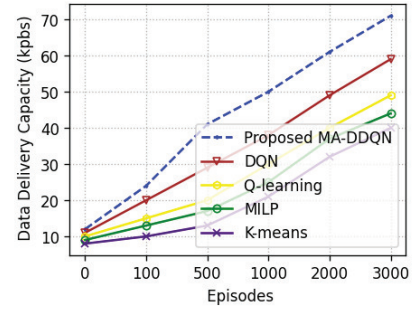
(a) Loss value Vs. number of Iteration.



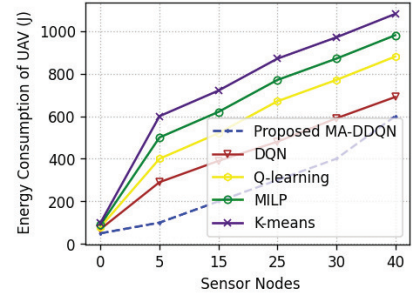
(b) Average reward Vs. number of Iteration.

Fig. 4. Convergence performance with a) loss value and b) memory batch size.

flights or data processing are needed to improve data quality. (A) Parameter Setting: In our simulation, optimizing UAV path planning and DCS for UAV-assisted RWSN is crucial to minimize task completion time and sensor node death time. Key parameters, such as data delivery, UAV energy consumption, task completion time, sensor lifetime, UAV charging rate, and sensor death time, are adjusted to enhance overall performance. (B) Training Process: Unlike greedy methods, the RL approach updates the Q-values of state-action pairs until convergence. The proposed MA-DDQN training process, involving typical average rewards and memory pool size, is outlined here. The trained module is then implemented and evaluated, as shown in Fig. 4.



(a) Data Delivery Capacity Vs. Episodes.



(b) Energy Consumption Vs. Sensor Nodes.

Fig. 5. Performance training includes: a) Data delivery capacity of the UAV relative to the episodes. b) Total energy consumption of the Sensor Nodes.

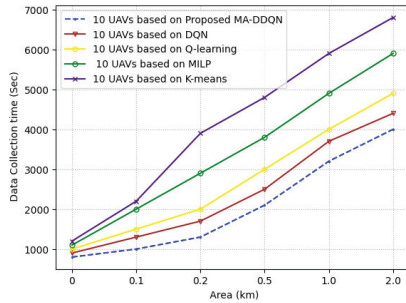
C. Data Delivery and Task Completion Analysis

Each simulation incorporates the DCS and trajectory path within the specified domain to minimize task completion time, and energy consumption, and maximize data delivery capacity in this scenario, as shown in Fig. 5.

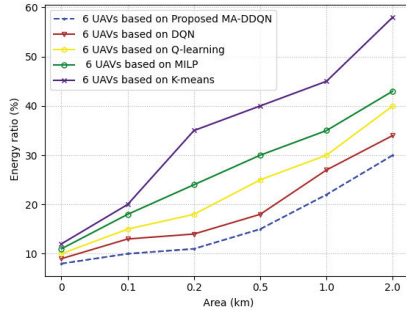
In multi-UAV RWSNs, the data delivery capacity of UAVs and the number of episodes increase during the data collection period for several reasons. Enhanced network coverage and shorter paths between sensors and the data sink enable UAVs to improve data delivery capacity, reducing latency and making data collection more efficient. Additionally, simultaneous data collection from multiple network parts, facilitated by deploying multiple UAVs, further enhances data delivery capacity, as shown in Fig. 5a. The increasing number of episodes can be attributed to the dynamic nature of the environment. To adapt to environmental changes, optimize data collection paths, or enhance overall network performance, a multi-UAV system may record more episodes during the data collection process. Higher communication and processing overheads arise from increased network traffic as the number of sensor nodes grows. This leads to greater energy consumption for data processing, transmission, and reception, as illustrated in Fig. 5b. Furthermore, with more sensor nodes, UAVs may need to fly longer distances or for extended durations to collect data, which demands additional energy from the UAV. Managing and coordinating a larger number of sensor nodes also consumes more power, further increasing overall energy consumption. Our proposed MA-DDQN method significantly outperforms other algorithms, achieving a performance of 87.25% compared to 67.85%, 55.25%, and 45.35% for

TABLE IV
SUMMARY OF TRAJECTORY PATH PARAMETERS (IN KM)

No.of UAVs	Proposed MA-DDQN	DQN	QL	MILP	K-Means
UAV1	13.2	19.2	20.2	23.4	25.5
UAV2	11.3	20.23	21.35	24.2	26.4
UAV3	10.14	17.45	18.5	20.3	23.5
UAV4	9.52	16.53	17.51	19.2	20.6
Total travel distance	44.16	57.41	77.56	87.1	96



(a) Data Collection Vs. Area



(b) Energy ratio Vs. Area

Fig. 6. Trajectory path planning: a) Data collection time with 10 UAVs b) Energy ratio with 6 UAVs.

the DQN, Q-learning, and K-means algorithms, respectively. This improvement demonstrates MA-DDQN's superior ability to increase data delivery efficiency while reducing energy consumption.

D. Trajectory Path Planning Analysis

To determine the number of SNs required for DCS in the RWSN, we propose to use the total travel distance of each UAV as shown in the Table. IV.

The increase in the data collection time and energy ratio is primarily due to the capabilities of UAVs that enable efficient data collection and deployment as shown in Fig. 6. UAVs can cover large areas quickly, using a parallel approach by multiple units, dramatically reducing the time required to collect data as shown in Fig. 6a. In addition, the low power

consumption associated with the latest UAV technologies supports extended operations despite the long distances involved in data collection. In addition, UAVs can act as mobile bases that facilitate continuous data transmission from deployed sensors, optimizing both time and energy consumption over large areas using multi-UAV RWSNs. In Fig. 6b, the main reasons for the energy ratio increase typically include factors like optimizing energy utilization through effective path planning, enhancing charging schemes to reduce energy consumption during operations, and maximizing terminal accessibility for data collection while navigating efficiently to recharge. An optimized trajectory can improve overall network performance by balancing the charging and operational demands of the UAVs.

The UAV trajectory path can increase with task completion time, the lifetime of sensors, and UAV charging energy in RWSN as elaborated in Fig. 7. The UAV trajectory path can increase with task completion time, lifetime of sensors, and UAV charging energy in RWSN by optimizing the flight trajectories to extend the operational time and efficiency of both the UAV and the network. By designing dynamic paths that prioritize charging zones and sensor coverage, operators can ensure that the UAV remains operational for longer periods, thus enhancing task completion and maximizing energy utilization, as indicated in studies focused on trajectory optimization for energy efficiency and effective charging management.

In Fig. 7a, as the task completion time increases, the UAV may need to cover a larger area to ensure all tasks are completed. This can result in a longer trajectory path for the UAV. The MA-DDQN method improves trajectory path efficiency by considering the collective actions of multiple agents, which leads to faster task completion compared to other DQN, Q-learning, MILP, and K-means. MA-DDQN learns a joint policy for all agents, allowing them to coordinate their movements and avoid collisions, resulting in smoother and more efficient paths. This approach outperforms methods that treat agents independently or rely on predefined heuristics, leading to more efficient use of resources and faster task completion, especially as the number of agents and the complexity of the task increase.

In Fig. 7b, as the sensor lifetime and the number of agents in the MA-DDQN method in multi-agent-based UAV-assisted

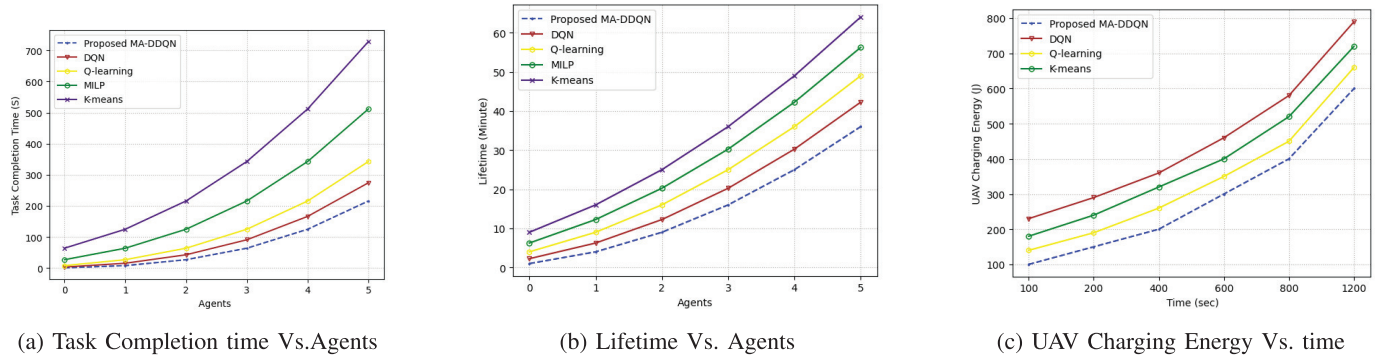


Fig. 7. Performance for Trajectory Path: (a) Task completion time for the number of agents (b) Lifetime of sensor nodes (c) UAV charging energy over time.

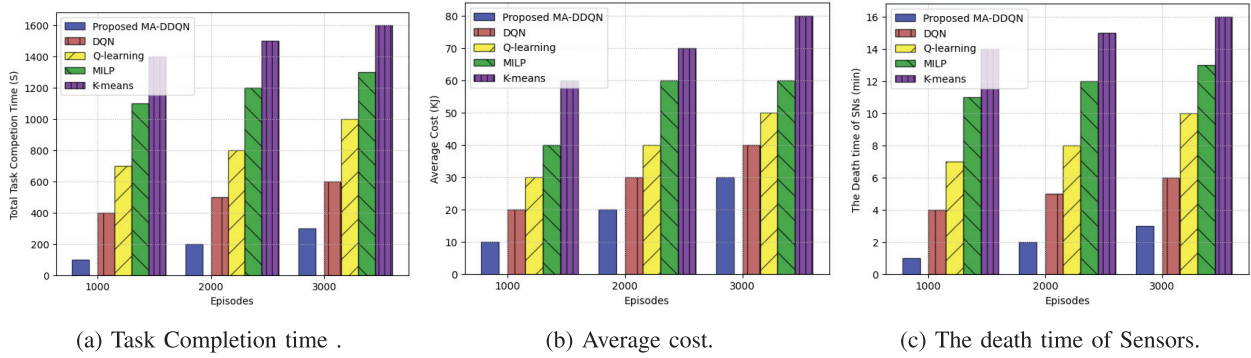


Fig. 8. The performance evaluation of the proposed method compared to conventional algorithms is presented in terms of: (a) Task completion time versus episodes (b) Average cost versus episodes (c) Death time of sensor nodes versus episodes.

WSN increase, the trajectory path improves in several ways: a) Improved coverage: With more agents and longer sensor lifetimes, the coverage of the area by the UAVs is improved. This means that a larger area can be monitored and maintained, resulting in a more efficient trajectory. b) Increased reliability: More agents and longer sensor lifetimes also provide redundancy in coverage, which increases reliability. If one agent or sensor fails, there are others to pick up the slack, ensuring continuous monitoring and service. c) Better resource allocation: With more agents and longer-lasting sensors, resources such as battery power can be allocated more effectively. Agents can take turns monitoring different areas or prioritize areas with sensors nearing the end of their life. d) Improved information gathering: More sensors with longer lifetimes means more data can be collected over time, leading to better analysis and decision-making to optimize trajectories.

Fig. 7c, the MA-DDQN method optimizes the UAV's trajectory path to maximize energy utilization. If the UAV has limited charging energy, it may need to conserve its energy and take more efficient paths between sensor nodes. However, if there are opportunities for recharging along its trajectory path, it may be able to cover more ground before needing to return for recharging. The proposed algorithm aims to find a path that balances the time spent charging the UAV with the time spent collecting data from the WSN. As the charging time increases, the UAV can fly for longer periods, enabling it to cover a larger area and collect data from more sensors. This leads to a more efficient and effective data

collection process, improving the overall performance of the WSN.

E. Energy Consumption, and Average Cost Analysis

To analyze our trajectory planning performance for rechargeable sensor nodes, we use MA-DDQN and compare our performance with other learning algorithms.

Fig. 8 illustrates how the increased energy level of the SNs can be attributed to the more efficient charging and coverage of the UAVs. The increased availability of UAVs will allow for more efficient charging of all SNs, higher energy levels, and longer network lifetimes through optimized trajectory patterns. The method we propose involves maximizing the energy levels of SNs by optimizing the trajectories of LCU for recharging purposes.

In Fig. 8a, we examine the MA-DDQN, DQN, Q-learning, and K-means performances to analyze the average cost, task completion time, and death time of SNs. As can be discovered, the proposed approach outperforms the other three benchmark algorithms as the number of episodes increases from 1000 to 2000 and the area increases in km. The performance of Fig. 8b (a) of the proposed method is improved by 89.35%, 77.35%, and 68.40% better than DQN, Q-learning, and K-means, respectively. The average cost grows as the number of episodes increases in Fig. 8a. The UAV-assisted rechargeable WSN's trajectory path simulates the system and gathers information on energy usage, flight duration, maintenance expenses, and other pertinent characteristics for varying

episode counts. We can see how the average cost varies with the number of episodes by examining this data and looking for any patterns or trends. Compared to other methods in this scenario, our proposed MA-DDQN can approximate the Q-values more accurately, resulting in the UAV making more accurate decisions when planning its trajectory for recharging the SNs. This could be an explanation for the increase in SN death time in Fig. 8c as the number of episodes is reduced. By making better decisions, the SN may use resources and energy more wisely, thereby increasing their lifetime.

VII. CONCLUSION

In this paper, we present an energy-efficient LCU-enabled RWSNs environment, where UAVs are deployed on the ground using laser beams to transmit energy to a base station to provide the services and collect the data from SNs in each time slot. We formulate the joint power, DCS, and path planning to minimize the task completion time and SN death time. The ultimate goal of using LCU-enabled RWSN technologies is to collect data at the lowest cost and with the shortest trajectory distance. We present a MA-DDQN technique to address this issue by using an MDP framework to obtain a dynamic charging strategy for DCS and different UAV trajectories. Our proposed MA-DDQN-based model was started to find the best solution for the charging SN strategy and determine the best possible trajectory for data collection in multi-UAV-assisted RWSNs. Furthermore, the energy consumption of RWSN nodes and UAVs during cooperative path planning can be a limitation, as it can reduce the overall lifetime of the network and the flight duration of the UAVs. In our future work, we intend to investigate the privacy and security issues associated with agricultural data transfer between UAVs and heterogeneous WSNs.

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